

Building a Backtesting Simulation for Trading Strategies

Nasdaq-100 Daily Data · April 2021 – April 2026

1. System Overview and Assumptions

This report describes the design, implementation, and experimental results of a daily close trading strategy backtesting simulation system built on five years of Nasdaq-100 constituent stock data (April 2021 – April 2026, 1,255 trading days, 101 stocks).

The system enforces the following assumptions throughout all experiments:

- **No lookahead bias:** strategies may only use price data up to and including the current trading day.
- **No short selling:** all portfolio weights are constrained to be non-negative.
- **No leverage:** the sum of all weights is at most 1.0; any remainder is held as cash.
- **No intraday trading:** all decisions and executions occur at the daily closing price.
- **Frictionless execution:** no transaction costs or slippage are modeled.
- **Normalized capital:** starting portfolio value is 1.0 for comparability across strategies.

2. System Architecture and Module Interaction

The system is organized into six modules that interact in a clean pipeline. Each module has a single responsibility, making it easy to extend or replace individual components without touching the rest of the codebase.

`data/loader.py`

Reads `nasdaq100_daily_5y.csv`, pivots from long format (ticker, date, close) to a wide price matrix (dates x tickers), and sorts by date.

`engine/backtester.py`

Core simulation loop. For each trading day t , calls `strategy.generate_weights(prices, t)`, enforces constraints, then computes portfolio return as the dot product of weights and next-day stock returns. Tracks NAV, daily returns, and weight history.

strategies/base.py

Defines `BaseStrategy`, an abstract class with a single required method `generate_weights(prices, current_date)`. All strategies subclass this. A `_no_lookahead()` helper slices history to the current date. Adding a new strategy requires no changes to any other module.

portfolio/constructor.py

Provides `uniform_weights()` (equal allocation) and `risk_adjusted_weights()` (inverse-volatility weighting), plus `rolling_volatility()` for computing annualized rolling vol.

metrics/performance.py

Computes cumulative return, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, win rate, and Calmar ratio from a `BacktestResult` object.

output/reporter.py

Saves metrics tables to CSV, plots NAV curves as PNG files, and appends timestamped experiment logs to a text file.

main.py

Entry point. Loads data, instantiates `Backtester`, imports strategies, runs backtests, and saves all outputs. Accepts `--deliverable {3,4,5,all}` to run specific experiment sets.

Module interaction flow:

```
main.py → data/loader.py → engine/backtester.py ↔ strategies/* →  
metrics/performance.py → output/reporter.py
```

3. Performance Metrics

The following metrics are used to evaluate all strategies. A strategy is considered strong if it achieves a high Sharpe ratio with a manageable maximum drawdown — raw returns alone are insufficient because higher returns often come with disproportionately higher risk.

Metric	Definition
Cumulative Return (%)	Total portfolio growth over the full period.
Annualized Return (%)	Geometric annualized return, assuming 252 trading days/year.
Annualized Volatility (%)	Std. dev. of daily returns scaled to annual.
Sharpe Ratio	Ann. return / ann. volatility (risk-free rate = 0). Above 1.0 is strong.
Maximum Drawdown (%)	Largest peak-to-trough NAV decline. Measures downside risk.
Win Rate (%)	Fraction of trading days with positive portfolio return.

Calmar Ratio	Ann. return / max drawdown . Return per unit of drawdown risk.
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4. Deliverable 3 — Single-Stock Strategy Evaluation (AAPL)

Five strategies were tested on Apple Inc. (AAPL) over the full five-year period (April 2021 – April 2026).

4.1 Strategy Descriptions

Momentum (lookback=20): Holds AAPL if its 20-day trailing return is positive; otherwise holds cash. Captures short-term price persistence.

Mean Reversion (lookback=20): Computes a 20-day rolling z-score of the price. Buys when z-score < -1.0 (price unusually low), expecting reversion to the mean.

MACD (12, 26, 9): MACD line = EMA(12) - EMA(26). Signal line = EMA(9) of MACD. Holds long when MACD is above the signal line (bullish crossover).

Bollinger Band (window=20): Buys when price drops below the lower band (mean - 2*std). Exits when price rises above the upper band (mean + 2*std).

RSI (period=14): Enters a long position when RSI falls below 30 (oversold). Exits when RSI exceeds 70 (overbought).

4.2 Results

Strategy	Cum. Return (%)	Ann. Return (%)	Ann. Vol (%)	Sharpe	Max DD (%)	Win Rate (%)	Calmar
Momentum (20d)	73.20	11.66	17.55	0.664	-19.75	29.88	0.590
Mean Reversion (20d)	42.78	7.41	16.04	0.462	-23.61	11.47	0.314
MACD (12,26,9)	57.50	9.55	17.02	0.561	-20.89	25.58	0.457
Bollinger Band (20d)	48.33	8.24	20.66	0.399	-30.22	22.79	0.273
RSI (14)	30.39	5.47	19.24	0.284	-30.22	20.80	0.181

4.3 Analysis

Momentum is the strongest performer on all risk-adjusted metrics (Sharpe 0.664, Calmar 0.590). This is consistent with the documented momentum effect in large-cap equities: stocks with recent positive price trends tend to continue in that direction over short horizons.

Mean Reversion achieves the lowest volatility (16.04%) but a lower Sharpe (0.462) and a very low win rate (11.47%), because it spends most of the time in cash waiting for dips that may not materialize in a trending bull market.

MACD performs between the two extremes, smoothing the momentum signal with EMAs to reduce whipsaw trades.

Bollinger Band and RSI are the weakest performers. Both are contrarian strategies that struggle in sustained uptrends: they repeatedly exit positions expecting a reversal, resulting in the highest drawdowns (-30.22%) despite the lowest returns. RSI's Sharpe of 0.284 highlights the pitfall of applying

mean-reversion logic to a structurally trending stock.

5. Deliverable 4 — Portfolio-Level Backtesting

5.1 Portfolio Construction Methods

Uniform Weighting: selected stocks receive equal allocation (1/N each). Simple and well-diversified, but ignores differences in risk across stocks.

Risk-Adjusted (Inverse Volatility) Weighting: each selected stock is weighted inversely proportional to its 20-day rolling annualized volatility. Stocks with lower volatility receive higher allocations, reducing the influence of highly volatile names.

5.2 Strategies Tested

SMA Portfolio: selects all stocks where the 20-day SMA is above the 50-day SMA (uptrend filter). Allocates equally across selected stocks.

TopK Momentum — Uniform: ranks all 101 stocks by 30-day trailing return, selects the top 10, and allocates equally.

TopK Momentum — Risk-Adjusted: same top-10 momentum selection, but uses inverse-volatility weighting instead of equal weight.

5.3 Results

Strategy	Cum. Return (%)	Ann. Return (%)	Ann. Vol (%)	Sharpe	Max DD (%)	Win Rate (%)	Calmar
SMA Portfolio (uniform)	72.00	11.51	18.97	0.606	-37.71	51.63	0.305
TopK Momentum (uniform, K=10)	239.09	27.79	27.59	1.007	-43.25	52.51	0.642
TopK Momentum (risk-adj, K=10)	158.27	20.99	25.43	0.825	-44.00	52.19	0.477

5.4 Analysis

TopK Momentum with uniform weighting achieves the highest cumulative return (239%) and the best Sharpe (1.007), demonstrating that cross-sectional momentum is a powerful signal in the Nasdaq-100. The top-10 stocks consistently captured the market's strongest performers (e.g., NVDA, META, MSTR during the AI-driven bull run of 2023–2025).

Risk-adjusted weighting reduces volatility from 27.59% to 25.43%, but also reduces returns more proportionally, resulting in a lower Sharpe (0.825). This occurs because the highest-momentum stocks in this dataset are often also the highest-volatility stocks — downweighting them reduces both risk and return simultaneously.

SMA Portfolio has a lower Sharpe (0.606) because the SMA crossover signal is slower to respond and selects a larger, more diluted set of stocks. Its max drawdown of -37.71% is lower than TopK strategies, making it more suitable for risk-averse portfolios.

6. Deliverable 5 — New Strategies vs Benchmark Strategies

The objective is to develop two new strategies that beat both benchmarks on Sharpe ratio. Benchmark 1 (SMA Portfolio) has Sharpe 0.606. Benchmark 2 (TopK Momentum, K=10) has Sharpe 1.007.

6.1 New Strategy 1: Risk-Adjusted Momentum with RSI Filter

Description: Computes 30-day trailing return for all 101 stocks. Filters out any stock with RSI above 70 (overbought — avoiding stocks vulnerable to reversals after large runs). From the remaining stocks, selects the top 15 by momentum and allocates using inverse-volatility weighting.

Rationale: The RSI filter adds a quality screen on top of momentum, avoiding crowded trades in overextended stocks. Inverse-vol weighting then allocates more to the smoother, steadier winners, improving the risk-adjusted return.

6.2 New Strategy 2: Market-Timed Momentum

Description: First checks the broad-market trend by computing the equal-weighted return of all 101 Nasdaq-100 stocks over the past 90 days. If this return is negative (broad market in a downtrend), the strategy holds 100% cash. Otherwise, selects the top 15 stocks by 30-day momentum and allocates using inverse-volatility weighting.

Rationale: Based on absolute momentum theory (Antonacci, 2012), a negative broad-market trend is a strong signal to exit risky assets entirely. This dramatically reduces drawdowns during market-wide selloffs (e.g., the 2022 rate-hike bear market) while staying invested during bull runs, yielding a much higher Sharpe ratio.

6.3 Results

Strategy	Ann. Return (%)	Ann. Vol (%)	Sharpe Ratio	Max DD (%)
Benchmark 1: SMA Portfolio	11.51	18.97	0.6064	-37.71
Benchmark 2: TopK Momentum (K=10)	27.79	27.59	1.0072	-43.25
New Strategy 1: RiskAdj Momentum	25.05	21.92	1.1431	-39.69
New Strategy 2: MarketTimed Momentum	24.70	18.23	1.3547	-11.84

Green rows = new strategies. Both exceed both benchmark Sharpe ratios.

6.4 Analysis

New Strategy 1 (RiskAdj Momentum, Sharpe 1.143) improves over TopK Momentum (1.007) primarily by reducing volatility from 27.59% to 21.92% through the RSI overbought filter and inverse-vol weighting. It avoids overextended names and concentrates capital in steady climbers.

New Strategy 2 (MarketTimed Momentum, Sharpe 1.355) achieves the best overall performance across all strategies tested. Its maximum drawdown of only -11.84% is dramatically lower than all other strategies — nearly 4x better than the TopK benchmark (-43.25%). The 90-day equal-weighted market filter successfully identified the 2022 bear market and moved the portfolio to cash, avoiding the worst of the drawdown. Annualized return of 24.70% is only slightly below the TopK benchmark (27.79%), but with 34% lower volatility (18.23% vs 27.59%), the risk-adjusted return is far superior.

7. Conclusion

This project delivers a complete, modular backtesting simulation system for Nasdaq-100 daily close strategies. Four key findings emerge:

1. **Momentum is the dominant signal.** All top-performing strategies use trailing return as their primary stock selection criterion. In the Nasdaq-100 over 2021–2026, momentum was consistently rewarded, particularly during the AI-driven bull market of 2023–2025.
2. **Risk management is as important as signal quality.** Strategies that combine momentum with risk controls (RSI filters, inverse-vol weighting, or market timing) achieve substantially better Sharpe ratios than pure momentum approaches.
3. **Market timing dramatically reduces drawdowns.** The 90-day broad-market filter in New Strategy 2 reduced maximum drawdown from -43% to -12% — nearly a 4x improvement — while preserving most of the upside return.
4. **Contrarian strategies underperform in trending markets.** Mean reversion, Bollinger Band, and RSI strategies all struggle in the sustained uptrend environment of this dataset.

The system is fully extensible — adding a new strategy requires only implementing `generate_weights()` in a new file in the `strategies/` directory, with no changes to the engine, metrics, or output modules.